

Question		Answer	Mark	Mark scheme	Additional guidance
14	(a)	Result shown	P1	for process to use the formula for the circumference of a circle, eg $2\pi r$ or $2\pi (r + t)$ or for process to use the sector angle, eg $\frac{108}{360}$ oe	Fractions may be simplified at any stage
			P1	for a complete process to find the perimeter, eg $\frac{108}{360} \{2\pi r + 2\pi (r + t)\} + 2t$	
			P1	for process to use a common denominator, eg $\frac{108 \{2\pi r + 2\pi (r + t)\} + 720t}{360}$	
			P1	for expansion of brackets and collection of terms, eg $432\pi r + 216\pi t + 720t$	
	(b)(i)	$6\pi + 20$	C1	for completion of processes to get the result.	Need to see steps in working, not only the final expression
	(ii)	explanation	B1	cao	
			C1	for explanation, eg this gives the perimeter of the sector (of circle, angle 108° , radius 10 cm)	